Image Mining, Practical Work

Part 1: content-based image retrieval on the Vertigo sequence

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1. Introduction

In this part I had to find, in the Vertigo dream scene, the frames that are related to a query image. Searching all the frames of the video would be long, so I first made a gallery of 150 frames, and then I tried different representations and similarity metrics on the four queries of the assignment. I started with a simple method each time, looked at what went wrong and used it to choose the next try, so the report follows the order in which I worked.

I used Python with PyAV to read the video, NumPy, Pillow, PyTorch and torchvision for ResNet-50, timm for DINOv2 (optional part) and Matplotlib for the figures. I ran the notebook on Google Colab and on my laptop, and the code chooses CUDA, MPS or CPU alone. To run it, the video zip and the portrait image must be in the same folder as the notebook.

1.1 Video and gallery

The video lasts 95 s, with 25 fps, so it has 2375 frames of 560×320 pixels. I took 150 frames evenly between 2 s after the start and 2 s before the end, which is one frame every 0.61 s or so. I checked that gallery image vertigo_020 is frame 356 at 14.24 s, like the assignment says.



I looked at the gallery before doing anything else, because it helps to understand the rankings later.

1.2 Queries

I used the four queries of the assignment: the whole frame 356, a crop of frame 356, a crop of frame 350 (which is not in the gallery), and the painting Portrait of a Lady. Both crops use the box (160, 40, 360, 240).



1.3 How I evaluate

All the methods use the same functions to rank the gallery (a higher score means more similar) and to do the four checks of the assignment. For the whole frame and the in-gallery crop, frame 356 must be first. For the held-out crop, frame 356 must be in the top 3. For the external image, one of the top-3 must have a frame index between 1184 and 1350. I also write down the rank of the first correct frame, because a fail at rank 2 is not the same as a fail at rank 50. In the figures a green border means the frame is correct for that query and a red border means it is not. The queries and the gallery always have the same preprocessing.

2. Colour histograms

2.1 First try: grey-level histogram

I started with the simplest thing, a histogram of the grey levels with 256 bins. I normalised it to sum to 1 because the frames and the crops don't have the same size, and I compared with cosine and with KL divergence.

Metric	Whole frame	In-gallery crop	Held-out crop	External
cosine	pass (rank 1)	fail (rank 40)	fail (rank 34)	fail (rank 4)
KL divergence	pass (rank 1)	fail (rank 19)	fail (rank 23)	fail (rank 24)



This didn't work. Only the whole frame is found, and that is the easy case because an identical image has an identical histogram. The two crops are matched with dark frames of the end of the film (around frame 2295), which are as bright as the crop but not blue. The external image is at rank 4 with cosine, which is close, but I think it is only luck. The problem is that the grey conversion removes the blue colour, and this colour is what makes this shot different from the others.

2.2 Second try: RGB histogram

Then I used a joint RGB histogram. Each channel is cut into 16 levels, so there are $16^3 = 4096$ bins, and a pixel is counted in the bin of its (R, G, B) values. I used a joint histogram because it keeps the colour combinations, for example dark blue and dark red go in different bins. I don't resize the images because a

histogram doesn't depend on the size. The 150 histograms are computed in 0.16 s. The Vertigo queries only use about 40 bins (dark blue, no red) and the painting uses about 290 bins of warm colours.

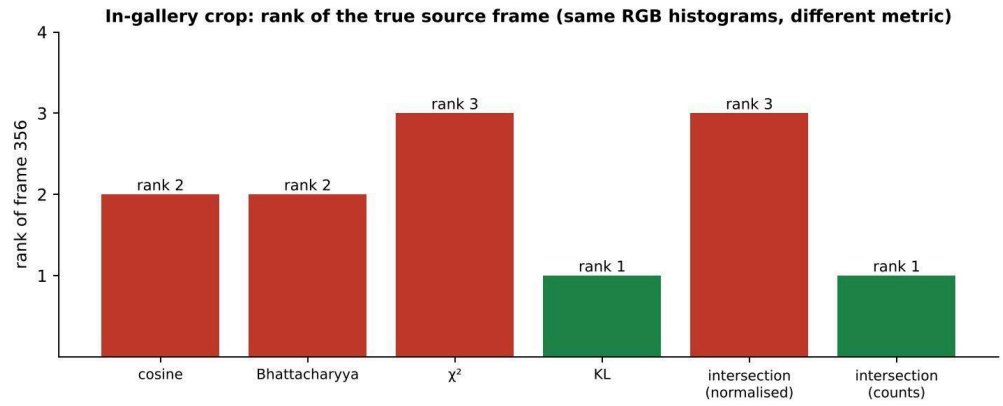
With cosine, the held-out crop is correct (rank 1), but the in-gallery crop is only second, and it is a piece of frame 356. This was strange to me, so I thought that now the representation is good and maybe the metric is the problem.

2.3 Third try: other metrics

I compared six metrics on the same RGB histograms. The last one is the histogram intersection of Swain and Ballard computed on the raw pixel counts and divided by the number of pixels of the query.

Rank of the first correct frame for each metric. A pass needs rank 1 for the first two queries and rank 3 or better for the last two.

Metric	Whole frame	In-gallery crop	Held-out crop	External
cosine	1	2	1	56
Bhattacharyya	1	2	1	27
chi-square	1	3	2	46
KL divergence	1	1	2	5
intersection (normalised)	1	3	2	48
intersection (raw counts)	1	1	1	54



Only KL and the intersection on raw counts put frame 356 first. My explanation is this. The normalised metrics check if the crop and the frame have the same colour distribution, and they don't, because the crop has no background, so any blue frame of the shot can win. The intersection on raw counts checks if the pixels of the query are inside the frame. If the frame has all of them the score is exactly 1, and that is the case of the in-gallery crop. I kept the intersection on raw counts because KL depends on the small value that is added for the empty bins.

2.4 Results of the colour histogram



Query	Result	First correct rank	Top-5 frames
Whole frame	pass	1	356, 432, 371, 447, 417
In-gallery crop	pass	1	356, 432, 371, 417, 447
Held-out crop	pass	1	356, 432, 371, 447, 417
External image	fail	54	1104, 1150, 1180, 1073, 1425

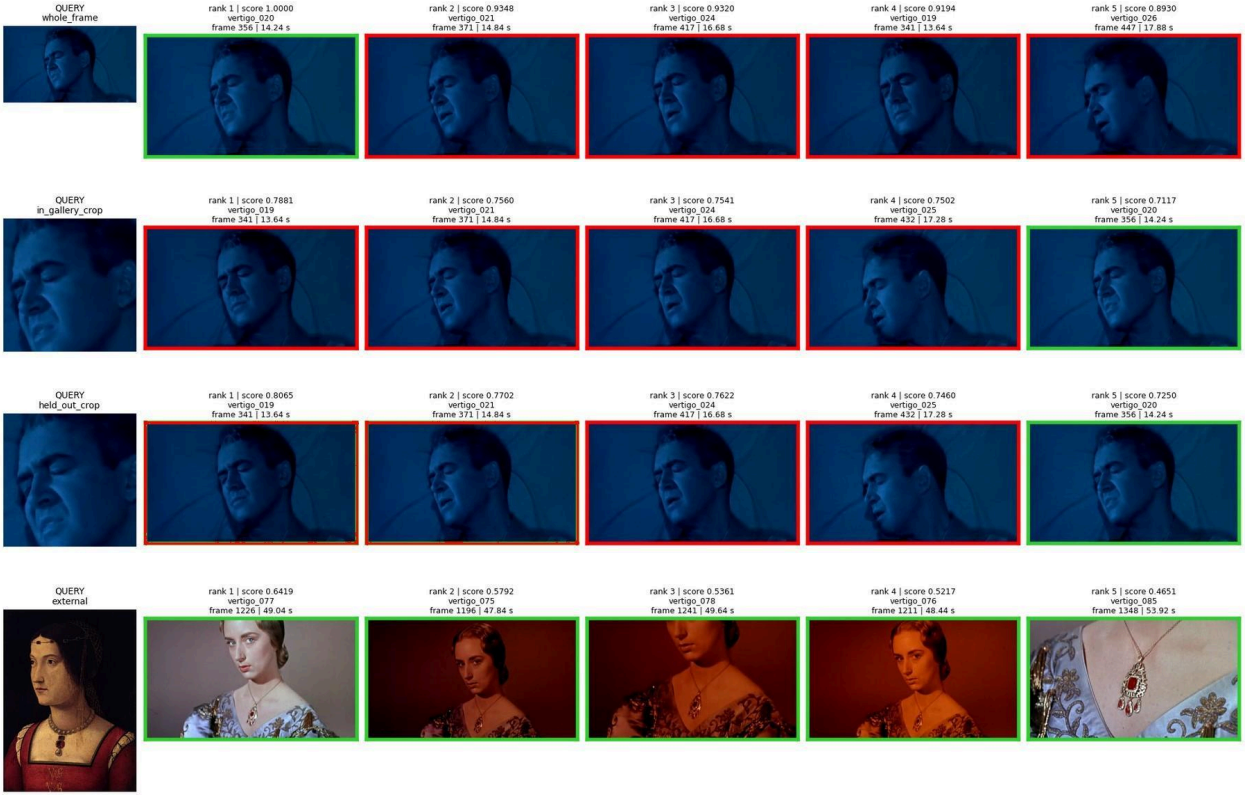
The histogram passes three checks out of four. Frame 356 has a score of exactly 1.0 for the whole frame and for the in-gallery crop, and for the held-out crop it finds frame 356 at 14.24 s, only 0.24 s from the query. The margin is small though: for the in-gallery crop the second frame (432) has 0.982, because the blue frames around it have almost the same pixels.

The external image fails. The first frame with the pendant is at rank 54, and the top results are the people at the window and a man in the dark. They have dark backgrounds and skin tones a bit like the painting but there is no pendant. A global histogram doesn't know anything about objects. I also think the good results on the crops are helped by the colour filter of the film, since only six frames are blue. In a video where all the frames have the same colours it would be worse.

3. ResNet-50

3.1 Global embedding

I used the torchvision ResNet-50 with the IMAGENET1K_V2 weights and I replaced the last layer with Identity, so I get a vector of 2048 values. The frames are converted from NumPy to PIL and then to a tensor of shape (batch, 3, 224, 224) with the official transform (resize to 232, centre crop to 224, ImageNet normalisation). Queries and gallery use the same transform. I compare with cosine similarity because the direction of the vector tells what is in the image, while its norm depends more on contrast. The 150 gallery frames take 1.2 s.



Query	Result	First correct rank	Top-5 frames
Whole frame	pass	1	356, 371, 417, 341, 447
In-gallery crop	fail	5	341, 371, 417, 432, 356
Held-out crop	fail	5	341, 371, 417, 432, 356
External image	pass	1	1226,1196, 1241, 1211, 1348

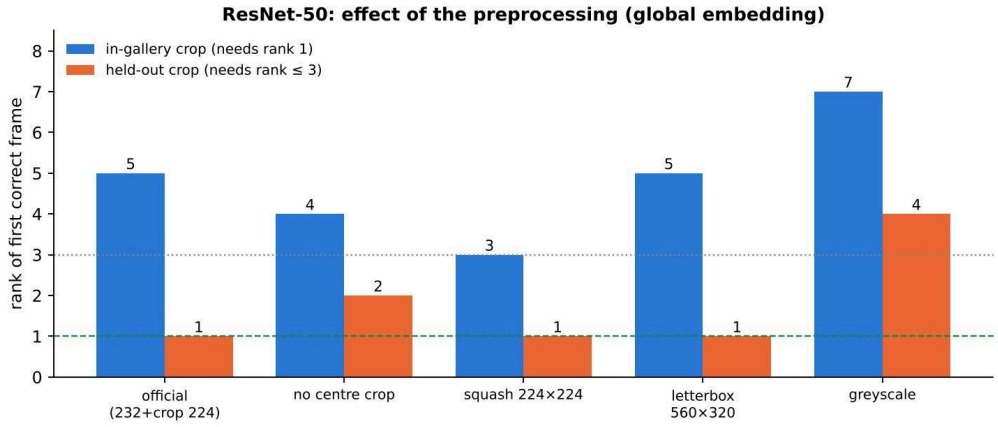
For the external query this is much better than the histogram. All five results show the woman with the pendant, even if the painting and the film don't have the same colours, so the network uses what is in the image and not only the colours. The held-out crop fails: frame 356 only appears at rank 5, behind four neighbours.

The in-gallery crop fails. Frame 356 is at rank 5, after four neighbours of the same shot. The answer is not wrong but it is not the exact frame. The in-gallery crop and the held-out crop have almost the same top-5, so the 10 grey levels of difference don't change anything in the embedding.

3.2 Changing the preprocessing

My first idea was that the centre crop is the problem. It removes about 45% of the width of the gallery frames, but the crops of the queries are square and are kept completely. So I tried five preprocessings, always the same for the queries and the gallery: the official one, no centre crop, resize to 224×224 without keeping the ratio, padding with black to 560×320 (letterbox), and greyscale.

Preprocessing	Whole frame	In-gallery crop	Held-out crop	External
official	1	5	5	1
no centre crop	1	4	4	1
resize 224×224	1	3	3	1
letterbox 560×320	1	5	5	1
greyscale	1	7	7	1

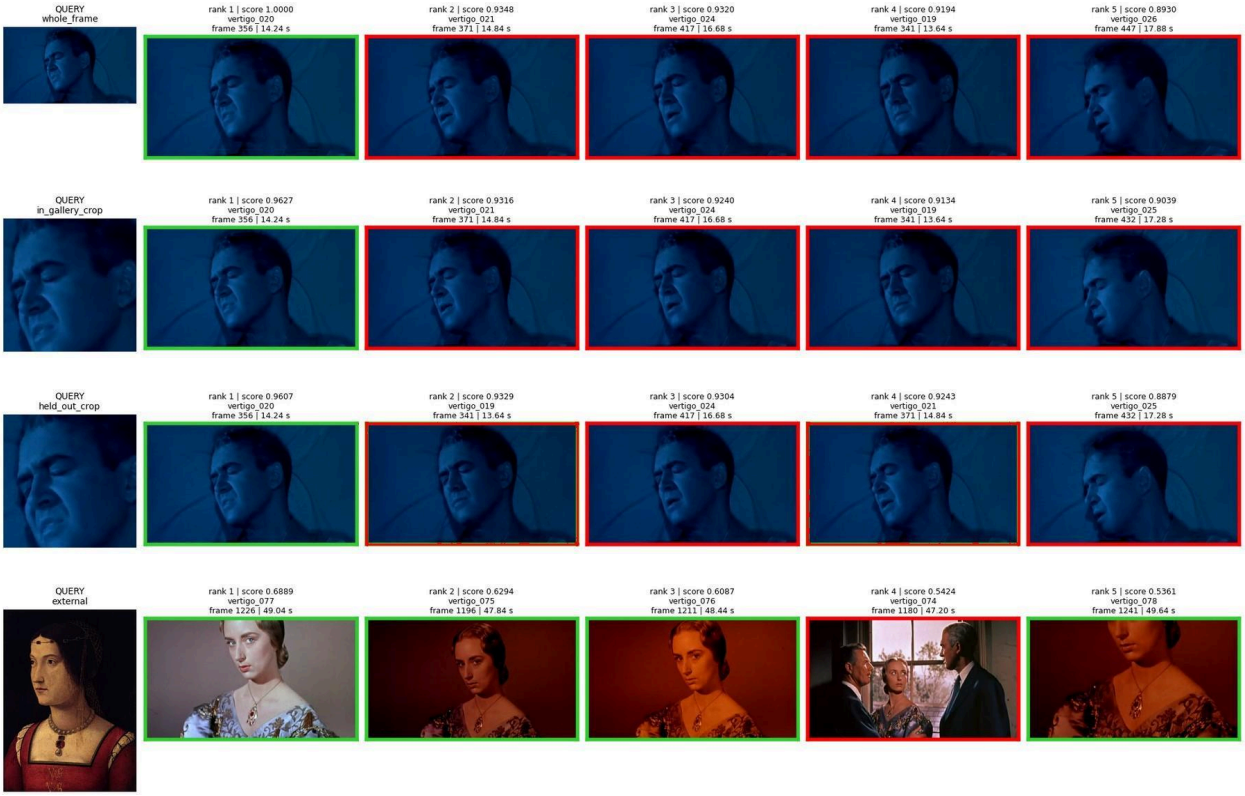


It was not enough. Without the centre crop or with the resize, frame 356 only goes from rank 5 to rank 4 or 3. Greyscale is the worst overall, like for the histogram, because the colour is useful in this video. The letterbox doesn't help because the black borders are not in the gallery frames. So I think the resizing is not the real problem. The embedding is global, and the crop is a big face while frame 356 is a smaller face in a big blue scene, so for the network these are two different images.

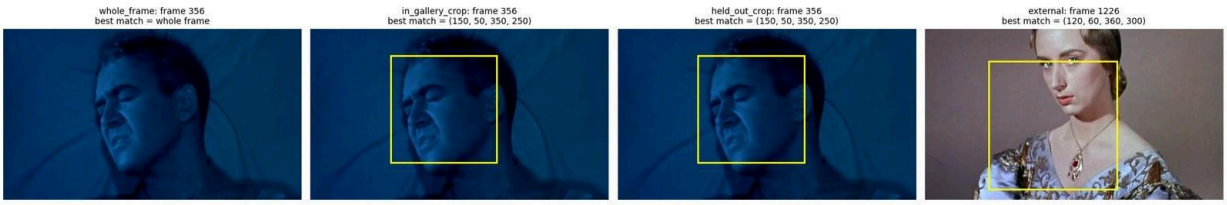
3.3 Regions

To solve this I used the same idea as for the histogram, which is to compare the query with parts of each gallery frame. For every frame I cut square windows of 160, 200 and 240 pixels, with a stride of a quarter of the window, which makes 91 windows per frame. Each window goes through the same ResNet-50 and the same preprocessing as the queries. A gallery frame is represented by its global vector and its 91 window vectors, and its score is the highest cosine similarity among them. I didn't put the real crop box

on the grid on purpose. The closest window is (150, 50, 350, 250), 10 pixels from the real box, so the method doesn't know where the crop comes from.



Query	Result	First correct rank	Top-5 frames
Whole frame	pass	1	356, 371, 417, 341, 447
In-gallery crop	pass	1	356, 371, 417, 341, 432
Held-out crop	pass	1	356, 341, 417, 371, 432
External image	pass	1	1226, 1196, 1211, 1180, 1241



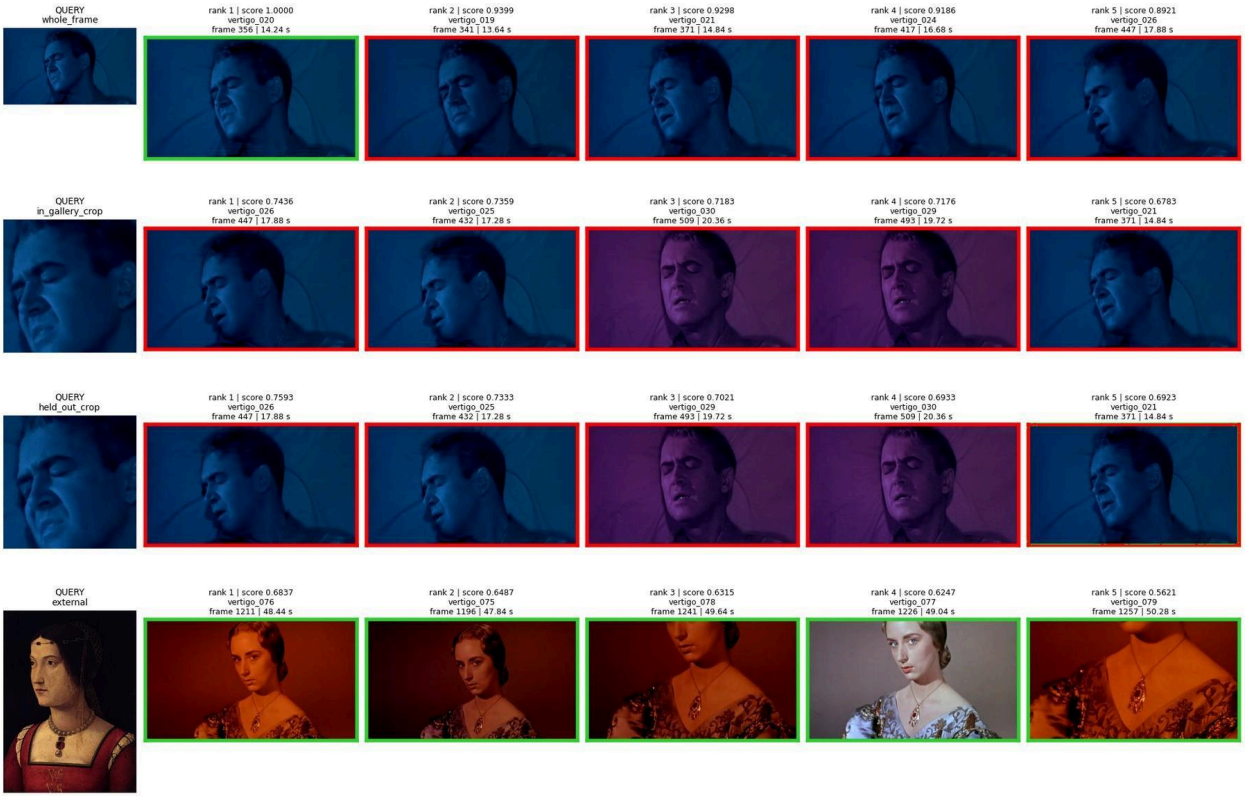
This time the four checks pass. For the in-gallery crop, frame 356 goes from rank 5 to rank 1, and the best window is on the face, near the real crop box. For the held-out crop frame 356 is also first now, and it is the closest gallery frame in time. For the external image, four frames of the top-5 show the pendant (frame 1180 is the window scene), and on frame 1226 the best window is the neck with the pendant, which is the object I wanted.

The problem is the cost. There are 92 vectors per frame instead of 1, and it takes 78 s instead of 1.2 s, about 65 times more. The scores are also higher in general because I take a maximum over many windows, so with a bigger gallery I would expect more false positives.

4. Optional part: DINOv2

My hypothesis was that DINOv2 is trained on a very large dataset without labels and is known to capture the meaning of the image better than a classification network. So I expected it to be at least as good as ResNet-50 on the painting, but not better on the crops, because the crops need one exact moment of the video.

I used DINOv2 ViT-S/14 from timm (21.6 M parameters, about the same as ResNet-50), with resize 256, centre crop 224, ImageNet normalisation, the class token of 384 values and cosine similarity. I didn't try bigger models because they are much slower and I don't think they would help for the crops.



Query	Result	First correct rank	Top-5 frames
Whole frame	pass	1	356, 341, 371, 417, 447
In-gallery crop	fail	7	447, 432, 509, 493, 371
Held-out crop	fail	7	447, 432, 493, 509, 371
External image	pass	1	1211, 1196, 1241, 1226, 1257

The results agree with the hypothesis. For the painting, all five results show the pendant, and it is the cleanest answer of all my methods. For the crops it fails. It gives frames 447 and 432 first, then 509 and

493, which are purple close-ups of the same man. So it knows it is the face of this man, but it doesn't care much about the blue colour or the exact moment, and these two checks need that.

5. Comparison

Method	Whole frame	In-gallery crop	Held-out crop	External	Passed
Colour histogram + intersection	1	1	1	54 (fail)	3/4
ResNet-50 global + cosine	1	5 (fail)	5 (fail)	1	2/4
ResNet-50 with regions + cosine	1	1	1	1	4/4
DINOv2 ViT-S/14 + cosine	1	7 (fail)	7 (fail)	1	2/4

I also compared the best score of a correct frame with the best score of a wrong frame on the two crops. The difference is +0.018 and +0.012 for the histogram, -0.076 and -0.082 for ResNet global, +0.031 and +0.028 for ResNet with regions, and -0.071 and -0.090 for DINOv2. All of them are small, because the wrong frames are neighbours of the same shot and they are almost the same image. Only the ResNet with regions is positive for both crops.

Some results surprised me:

- the grey histogram matches the crops with the end of the film, because it has the same brightness but the blue is lost;
- the normalised colour metrics don't put frame 356 first for its own crop, but the intersection on raw counts does, with exactly 1;
- the two crops have the same top-5 with ResNet, so the network doesn't see the small change between frames 350 and 356;
- ResNet global puts four neighbours before frame 356, I think because it sees the frames of a shot as almost the same and the crop has another scale;
- DINOv2 returns purple close-ups before blue ones, because it uses the meaning of the image and not the colour filter;
- for the painting, the histogram returns the people at the window, because of the dark background and the skin colours.

6. Conclusion

The colour histogram with the intersection on raw counts finds the exact source frame of a crop with a score of 1.0, quickly and without any learning. ResNet-50 and DINOv2 find the frames with the pendant from a painting. The only method that passes the four checks is ResNet-50 with regions.

What didn't work: the grey histogram, because it loses the colour; the normalised colour metrics on the crops; the histogram on the painting (rank 54); ResNet global on the in-gallery crop (rank 5) and on the held-out crop (also rank 5); and DINOv2 on both crops.

What I learned is that the metric has to fit the kind of query. With the same histograms, the metrics that compare distributions failed on a crop but the intersection on raw counts worked. Colour is useful inside one video but it doesn't give any meaning, while deep features know what is in the image but they see neighbouring frames as the same. Changing the preprocessing didn't solve the crop, but changing what I compare (the query against the best region of the frame) did, even if it needs about 90 times more vectors. Using the same preprocessing for queries and gallery is needed, but it is not enough when the query is another kind of image than the gallery images.